Padraig Lavery

Software Engineer – Production Reliability & Incident Response

Belfast, Northern Ireland | LinkedIn | GitHub

PROFILE

Software engineer with 4+ years building and running high-volume, security-sensitive production services for UK and Middle East banking clients (Node.js, .NET/C#, PostgreSQL on AWS and Kubernetes). Go-to engineer for production incidents: finds the root cause, fixes it permanently, and leaves runbooks so it doesn't recur. Operates what the team builds, from fix through release. Building Incident Copilot, an incident-triage service using retrieval and agentic tool-calling.

EXPERIENCE

Developer | Reward, Belfast | June 2022 – Present

Incident ownership and production fixes

- Resolved a live customer-facing outage caused by malformed load-balancer group naming in a Kubernetes/Istio ingress setup, then corrected the Helm deployment templates so it could not recur.
- Eliminated recurring daily out-of-memory crashes in a high-volume transaction ingestion service by re-architecting cache loading to run once per batch instead of per file; diagnosed and mitigated further memory-exhaustion failures in ETL jobs.
- Redesigned a month-end transactional invoicing batch with per-partner error isolation, explicit timeouts and safe retry/reconciliation, so failures on the highest-volume partners no longer abort the whole run.
- Authored runbooks and SOPs (re-sending failed outbound files, re-running failed jobs) so the production services team resolves recurring issues without engineering escalation.

Platform, automation and security

- Automated scheduled file processing with EventBridge, Lambda and Fargate, removing manual triggers; audited 100+ S3 buckets and applied lifecycle policies.
- Authored the engineering guide for upgrading AWS SDK for JavaScript v2 to v3 across Lambdas, Fargate services and microservices, and planned testing for the SDK and Node 22 Lambda runtime upgrades across repositories and bank deployments.
- Hardened authentication and account services: removed stack traces and SQL details from error responses and standardised input validation across partner microservices; remediated critical Snyk-flagged vulnerabilities for PCI DSS compliance.
- Designed and built audit logging of staff access to customer accounts across UK and Middle East bank portals to meet client security audit requirements.

Delivery and leadership

- Led a cross-team initiative to enable automated API testing: request validation, shared error handling, safe test-data cleanup endpoints, OpenAPI documentation, unit tests, and a stabilised GitHub Actions CI pipeline.
- Mentored and onboarded four new starters; run the team's retrospectives and author technical guides.

Intern Software Engineer | Flexera, Belfast | May 2020 – April 2021

- Developed a web API for calculating cloud costs in a small agile team using Go, Docker, Kubernetes and Git.

PROJECTS

Incident Copilot (in progress) | Python, FastAPI, PostgreSQL + pgvector, Ollama, Docker, GitHub Actions

- Incident-triage service: takes a production incident description, retrieves matching runbook content and suggests likely root causes and next steps, with agentic tool-calling for live service status. Runbook knowledge base and FastAPI service in place; retrieval and model integration in development.

SKILLS

- **Operations:** production incident response, root cause analysis, runbooks/SOPs, security remediation (Snyk, OWASP)
- **Languages:** JavaScript (Node.js), C#, Python, Go, SQL
- **Cloud and infrastructure:** AWS (Lambda, Fargate, S3, EventBridge), Kubernetes, Helm, Istio, Docker, Terraform, GitHub Actions
- **Back end and data:** Node.js, .NET Core, FastAPI, microservices, OpenAPI, PostgreSQL, MongoDB
- **AI:** retrieval-augmented generation, agentic tool-calling, local LLMs (Ollama)

EDUCATION AND CERTIFICATIONS

- BSc (Hons) Computer Science, 2:1 – Ulster University, Belfast, June 2022
- AWS Certified Developer – Associate; AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner – preparing, exam November 2026